

Exercise 2.1. Suppose n unrelated people are gathered together. What is the smallest n for which chances are $>50\%$ that there will be two or more people born in the same calendar month?
"At least one": Use Complement

Find the smallest n such that: $P(\text{At least two people born in the same month}) \geq 0.5$

(X) Direct Calculation?

- $P(\text{two people share a month})$
 - $P(\text{exactly three})$
 - $P(\text{four people share same month})$
 - $P(\text{two pairs in different months}) \dots$
- Too many ways ...

(✓) Complement Calculation?

- $1 - P(\text{No one shares same month}) = P(\text{At least one match})$
"n" people all have different months

- Calculating $P(\text{No one shares same month})$:

P'	$1 - P'$
$n=1: P: \frac{12}{12} = 1$	0 (0%)
$n=2: P: \frac{12}{12} \cdot \frac{11}{12} \approx 0.9167$	≈ 0.0833 (8.33%)
$n=3: P: \frac{12}{12} \cdot \frac{11}{12} \cdot \frac{10}{12} \approx 0.7639$	≈ 0.2361 (23.61%)
$n=4: P: \frac{12}{12} \cdot \frac{11}{12} \cdot \frac{10}{12} \cdot \frac{9}{12} \approx 0.5729$	≈ 0.4271 (42.7%)
$n=5: P: \frac{12}{12} \cdot \frac{11}{12} \cdot \frac{10}{12} \cdot \frac{9}{12} \cdot \frac{8}{12} = 0.3819$	≈ 0.6181 (61.8%)

$n=5$

The smallest number of people such that the chance of at least two sharing the same birth month exceeds 50%

Exercise 3.1. A fair die is rolled twice. What is the probability that at least one of the two rolls is a six if the sum of the two rolls is at least 10?

$P(B) = \text{All Possible Combinations} : 6$

Two rolls w/ sum ≥ 10

- (6, 4)
 - (4, 6)
 - (6, 5)
 - (5, 6)
 - (6, 6)
- (5, 5) $\times \rightarrow$ Doesn't contain 6
- $P(A \cap B) = 5$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = 5/6$$

- A: at least one of rolls a '6'.
- B: the sum of two rolls ≥ 10 .

Exercise 3.4. Suppose $P(A) = P(B) = .9$. Give a useful lower bound on $P(B|A)$.

Inclusion-Exclusion Formula
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Conditional Probability
 $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq 1$$

↓ rearrange

$$P(A \cap B) \geq P(A) + P(B) - 1$$

flip sign

Plug in $\circ P(A) = 0.9$
 $\circ P(B) = 0.9$

$$P(A \cap B) \geq 0.9 + 0.9 - 1$$

$$P(A \cap B) \geq 0.8$$

- No matter what, the p (both events happening) can't be lower than 0.8.
- Overlap is at least 0.8.

Apply to Conditional Probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\geq 0.8}{0.9}$$

$$P(B|A) \geq \frac{8}{9} \quad \text{Lower Bound for } P(B|A)$$

Exercise 3.10. Three independent proofreaders will read a manuscript. The probabilities that a particular error will be detected by them are respectively .92, .85, and .95. What is the probability that the error will go undetected?

$$\begin{aligned} A_1 &: \{ \text{Proofreader 1 detecting error} \} \\ A_2 &: \{ \text{Proofreader 2 detecting error} \} \\ A_3 &: \{ \text{Proofreader 3 detecting error} \} \end{aligned} \Rightarrow P(A_1 \cap A_2 \cap A_3) \rightarrow \text{All proofreaders detecting error}$$

$$\begin{aligned} A_1^c &: \{ \text{Proofreader 1 misses error} \} \\ A_2^c &: \{ \text{Proofreader 2 misses error} \} \\ A_3^c &: \{ \text{Proofreader 3 misses error} \} \end{aligned} \Rightarrow \begin{aligned} & \text{⊗ } P(A_1^c \cap A_2^c \cap A_3^c) \neq 1 - P(A_1 \cap A_2 \cap A_3) \rightarrow \text{At least one misses error.} \\ & \text{⊙ } P(A_1^c \cap A_2^c \cap A_3^c) = [1 - P(A_1)] \cdot [1 - P(A_2)] \cdot [1 - P(A_3)] \end{aligned}$$

$$P(A_1^c \cap A_2^c \cap A_3^c) = (1 - 0.92)(1 - 0.85)(1 - 0.95)$$

$$P(A_1^c \cap A_2^c \cap A_3^c) = 0.0006$$

Probability all proofreaders miss error

Exercise 3.11. * Two numbers are chosen without replacement from $1, 2, \dots, 100$.

What is the probability that the smaller number is greater than 20 if the larger number is smaller than 60?

Conditioned on larger number

Let: X be smaller number
 y be larger number

• Possible X and y ranges to satisfy $X < y$:

$X: \{1 \dots 99\}$

$y: \{2 \dots 100\}$ Total of 99 possible values each.

• Possible X ranges to satisfy $X > 20$:

$X: \{21 \dots 99\}$

Total of 79 possible values.

$X < y < 60$

• Possible y ranges to satisfy $y < 60$:

$y: \{2 \dots 59\}$

Total of 58 possible values. (for y only)

Accounting for X , additional $X=1$

Total of 59 possible values $X < y < 60$

• Count $\binom{59}{2} = 1711$

$20 < X < y < 60$

• Possible X and y ranges to satisfy $X > 20 \cap y < 60$:

$X: \{21 \dots 99\}$

$y: \{2 \dots 59\} \Rightarrow$ Overlapping: $\{21 \dots 59\}$

Total of 39 possible values. $20 < X < y < 60$

• Count: $\binom{39}{2} = \frac{39 \cdot 38}{2} = 741$

Combination: order is not important

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

★ **Equilikely Outcomes:** In an equally finite sample space, the probability:

$$P(E) = \text{Favorable Outcomes Count} / \text{Total Outcomes Count}$$

$$P(X > 20 | y < 60) = \frac{P(X > 20 \cap y < 60)}{P(y < 60)}$$

$$P(X > 20 | y < 60) = \frac{\overset{\text{count}}{|X > 20 \cap y < 60|} / \binom{100}{2}}{|y < 60| / \binom{100}{2}}$$

$\Downarrow$

$$P(X > 20 | y < 60) = \frac{\text{Count of } X > 20 \cap y < 60}{\text{number of } (x, y) \text{ pairs}} / \text{Count of } y < 60$$

$$P(X > 20 | y < 60) = 741 / 1711 = 0.4332$$

Homework 2

Stat-516

Problem 2.1

Let n be the number of people. We want the smallest n such that the probability of at least two people being born in the same month exceeds 0.5. It is easier to compute the complement probability: that all n birthdays fall in different months. For $n \leq 12$, this probability is

$$P(\text{all different months}) = \frac{12 \cdot 11 \cdot 10 \cdots (12 - n + 1)}{12^n}.$$

$$P(\text{at least one shared month}) = 1 - \frac{12 \cdot 11 \cdots (12 - n + 1)}{12^n}.$$

For $n = 4$,

$$P(\text{shared month}) = 1 - \frac{12 \cdot 11 \cdot 10 \cdot 9}{12^4} = 1 - \frac{11 \cdot 10 \cdot 9}{12^3} \approx 0.42708 < 0.5.$$

For $n = 5$,

$$P(\text{shared month}) = 1 - \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{12^5} = 1 - \frac{95040}{248832} \approx 0.61806 > 0.5.$$

Therefore, the smallest n such that the probability exceeds 50% is $\boxed{5}$.

Problem 3.1

Possibilities in two rolls sum is at least 10 = $\{(5, 5), (4, 6), (6, 4), (6, 5), (5, 6), (6, 6)\}$ So the number of such possibilities is 6. Notice that all of these have one 6 in the rolls except $(5, 5)$. So the probability we seek is $\frac{5}{6}$.

Problem 3.4

$P(A \cup B) \leq 1$. This is the trivial bound that has to be followed.

Expanding we get $P(A) + P(B) - P(AB) \leq 1$.

As $P(A) = P(B) = 0.9$ we get the bound on $P(AB) \geq 0.8$ (1)

$P(B|A) = \frac{P(AB)}{P(A)}$, Using the bound of $P(AB)$ we get $P(B|A) \geq \frac{0.8}{0.9}$.

Problem 3.10

let $P(A)$ = Probability error detected by proof reader 1; Similarly $P(B), P(C)$ are probabilities of error detected by proof reader 2 and 3.

Probability no error is detected = $1 - P(A \cup B \cup C)$

= $1 - [P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC)]$

As the proof readers are independent the events A, B, C are independent. Hence all intersection of events decompose into products in terms of probabilities.

= $1 - [P(A) + P(B) + P(C) - P(A)P(B) - P(B)P(C) - P(A)P(C) + P(A)P(B)P(C)]$

substitute values of $P(A), P(B), P(C)$ as 0.92, 0.85, 0.95 given in the problem to get the solution 0.0006.

Problem 3.11

Given larger number is less than 60, is equivalent to saying both numbers lie between 1–59. Total no. of such possibilities = $\binom{59}{2}$. (choosing two numbers between 59 numbers)

Now among these possibilities we want smaller number to be greater than 20, that is both numbers should be in the range 21–59. So 39 numbers to choose from. No. of ways = $\binom{39}{2}$.

Probability of such an event is then the ratio = $\frac{\binom{39}{2}}{\binom{59}{2}}$.